



KING FAHD UNIVERSITY OF PETROLEUM & MINERALS

College of Computing and Mathematics

Information and Computer Science Department

ICS 483: Computer Vision (3-0-3)

Second Semester (2024-2025) Term 242

PROJECT INSTRUCTIONS

Overview: This is an opportunity for you to apply the learnings from this class to a problem of your interest.

The potential projects can fall into the following two categories:

1. Applications. If you are interested in a particular area of study (e.g., engineering, medical, etc.), it will be very interesting to see how you apply computer vision to problems related to that area of your interest. You can select a real-world problem and apply the computer vision techniques we have covered in the course to solve it.
2. Models. You can also build a new technique or algorithm or modify and improve an existing model for computer vision and apply it to tackle some real-world computer vision tasks. This category, however, might be more challenging than the first one.

Additional Details:

The following criterion must be met in your project:

1. If you have selected the project of type ("Applications"), it should be solving a real-world problem.
2. The dataset for your selected problem must be publicly available.
3. You need to perform some preprocessing tasks on your dataset before you pass it on to the next stage of your solution:
 - a. It could be "Image Enhancement" or other preprocessing tasks.
 - b. You need to compare the results of your solution "with" the preprocessed dataset and "without" preprocessed dataset.
4. After preprocessing the dataset, you can move to the next phase of your solution, the actual task that you want to perform for your dataset. For example, recognition.

5. To perform a recognition task, you need to select statistical features (e.g., using SIFT, HOG, and other techniques available in the literature). Once you have extracted the statistical features, you can:
 - Do the classification using, for example, SVM model.
 - Use these features with Deep Learning models.
6. Run the same technique you have selected in the previous step and run it on raw images as well.
7. Compare the results from the technique for raw images (Step 6) and enhanced images (Step 5).

Project Proposal

A one-page document summarizing your selected problem for the project, the motivation behind selecting that problem, the availability of datasets, and the solution you plan to implement. Please also include the group members' information who are participating in the project.

Note: The page limit for the project proposal is strictly 1-2 pages (PDF). A group must be of 4 students.

Project Report

You will be submitting a project report (PDF) towards the end of the course. The instructions for the project report can be found on the next page.

Project Presentation/Demo

You also need to give a short presentation of your project along with a demo where you will briefly talk about the problem statement, the solution that you have implemented, and the results. There will also be a Q/A session at the end.

PROJECT REPORT INSTRUCTIONS

- Use IEEE Research Paper Template (**10 points**)
- A detailed introduction (this is w.r.t. to your selected problem) (**10 points**)
- Motivation (this is the motivation behind selecting this problem: why did you select this problem? and that how your solution could be helpful, and who could be the target audience to use it?) (**5 points**)
- Selected solution (This is where you will briefly describe which algorithms and techniques you have selected to address the selected problem) (**5 points**)
- Methodology (this is where you will explain all the steps you have taken to solve the problem. In other words, this is the complete pipeline of your solution. Here you will also explain, for example, if you have performed any image preprocessing tasks and image enhancement tasks. Make sure that there is proper reasoning that why are you performing any specific task) (**20 points**)
- Diagrams/Flowcharts (you need to have an overview diagram of your solution. Make sure that this diagram conveys your whole solution) (**10 points**)
- Dataset(s) (this is where you will include statistics about your dataset. You need to include charts and tables to showcase the details of your dataset(s). Please also include a few samples from the dataset as well.) (**10 points**)
- Experiment and Results (this is where you will explain what experiment did you run, how the model was trained, how many samples did you include in the training, and what modifications in the model you have done to increase the efficiency and accuracy of the model, how the model is evaluated, make sure that there is a reasoning for any decision/step you have taken or performed, you will include accuracy charts for your models, you will show unknown samples you have used for the evaluations, you will also include the graphical output of your experiments, for example, tagging that is done by your model to identify something) (**20 points**)
- Discussion (this is your opinion on the results. You will add your own thoughts and insights about the results you are getting, why you think you are getting good or bad results, what can be done to improve it, and any other thing that you want to report) (**10 points**)